



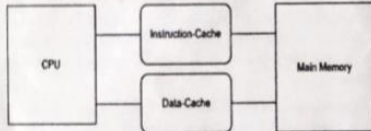
VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – June 2023

Programme	: B.Tech (ECM)	Semester	: Fall Inter 2022-23
Course	: Computer Architecture and Organization	Code	: BCSE205L
		Slot	: A2+TA2
Faculty	: Dr. Sindhuja M	Class Nbr.	: CH2022232500336
	: Dr. Sunil Kumar Pradhan		: CH2022232500337
	: Dr. Balakrishnan R		: CH2022232500338
	: Mr. Ajeyprasaath Kb		: CH2022232500340
Time	: 90 Minutes	Max. Marks	: 50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks
		Consider a 2-level memory hierarchy with separate instruction and data caches in level 1, and main memory in level 2. Figure 1 for the reference.	10
			
		Figure 1	
1.		The following parameters are given: a) The clock cycle time is 2ns b) The miss penalty is 12 clock cycles (for both read and write) c) 2% of instructions are not found in instruction cache d) 5% of data references are not found in data cache e) 25% of the total memory accesses are for data f) Cache access time (including hit detection) is 1 clock cycle Find average access time.	
2.		How many 256 x 4 RAM chips are needed to construct a memory chip of 2048x8? Show the corresponding interconnection diagram.	10



Consider a direct mapped cache of size 4 KB with block size 1024 bytes. The size of main memory is 1MB. Find-

3.
 1. Number of Bits in Physical Address [2 marks]
 2. Number of Bits in Block Offset [2 marks]
 3. Number of bits in tag [3 marks]
 4. Tag directory size [3 marks]

- (a) On a non-pipelined sequential processor, the following program segment, that is part of the Interrupt Service Routine (ISR), is given to transfer 500 bytes from an I/O device to memory.

Initialize the memory address register
Initialize word count register to 500
Loop: Load a byte from device
Store in memory at address given by address register
Increment the memory address register
Decrement the word count register
If count != 0 goto Loop

4. Assume that each statement in the program is equivalent to a machine instruction that takes 1 clock cycle to execute if it is a non LOAD/STORE instruction. The LOAD/STORE instructions take 2 clock cycles to execute. The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes 2 clock cycles to transfer one byte of data from the device to the memory.
 What is the approximate speedup when the DMA controller-based design is used in place of the interrupt driven approach? [8 marks]

- (b) Suppose that a disk is rotating at a speed of 10,000 rpm, and there are 120 Kbytes of data recorded in every track. Once the disk head reaches the desired track, what is the sustained data transfer rate in Mbytes/sec? [2 marks]

The size of the word count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 Kbytes from disk to main memory. The memory is byte addressable. What is the minimum number of times the DMA controller needs to get control of the system bus from the processor to transfer the file from disk to main memory? [5 marks]

5. Assume 1 kb = 1024 bytes.

- (a) A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much % will the CPU be slowed down because of the DMA transfer? [5 marks]

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Continuous Assessment Test - II, April 2024

Winter Semester 2023-2024

Course Name, Code:

Programme:

Class Numbers:

Slot: D1+TD1

Faculty:

Duration: 90 mins

Max Marks: 50

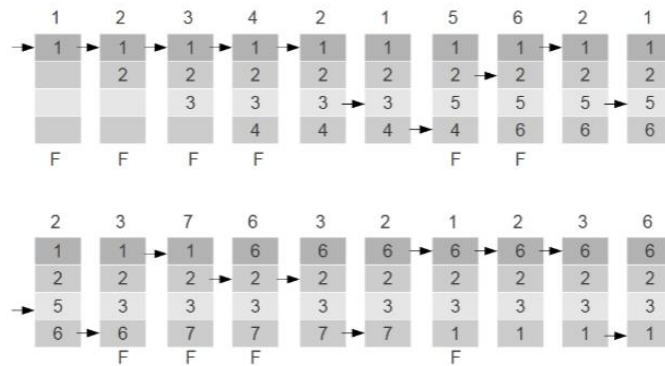
General instruction(s): Write question number properly on answer scripts.

Sl. No	Question
1	a. $CPI = \sum_i (\text{Fraction of instruction type } i \times \text{Cycles needed by instruction type } i)$ $\Rightarrow CPI = 0.45 * 1 + (0.32 + 0.08 + 0.15) * 2 = 1.55$ Execution time = CPI * total number of instructions * Clock time $= 1.55 * 10^5 * \frac{1}{40 * 10^6} \text{ sec}$ $= 3.875 \text{ ms}$ $MIPS = \frac{\text{Clock Frequency}}{CPI * 10^6}$ $= \frac{40 * 10^6}{1.55 * 10^6}$ $= 25.8$ b. Micro programmed control is a control mechanism to generate control signals by using a memory called control storage (CS), which contains the control signals. Although micro programmed control seems to be advantageous to CISC machines, since CISC requires systematic development of sophisticated control signals, there is no intrinsic difference between these 2 control mechanisms. Hardwired control is a control mechanism to generate control signals by using appropriate finite state machine (FSM). The pair of “microinstruction-register” and “control storage address register” can be regarded as a “state register” for the hardwired control. Note that the control storage can be regarded as a kind of combinational logic circuit. We can assign any 0, 1 values to each output corresponding to each address, which can be regarded as the input for a combinational logic circuit.

2	Suppose that a $2M \times 16$ RAM memory is built using $256K \times 8$ RAM chips and $1K \times 8$ ROM memory is built using 256×8 ROM chips with the word addressable memory, find the following: a) 16 RAMS , 4 ROMS b) Each RAM chip is $256K \times 8$, so to access one full word (2 bytes), 2 RAM chips would be involved. c) The $256K \times 8$ RAM chip has 18 address bits ($2^{18} = 256K$), so each RAM chip would need 18 address bits. d)8 e)module 0 – word 27 module 3 - word 3				
3	a. 16 Kb cache size. 4 way set associative. Line size = $4 \times 32 \text{ bit} = 16 \text{ bytes}$. Here, in question "word" is mentioned and even <i>"Where in the cache is the word from memory location"</i> is asked. So, word addressing is in use. So, offset bits = 2 for 4 words. No. of sets=no of lines/p(way) No. of lines=cache size /line size $\#sets = \frac{2^{10}}{4} = 256$ <table border="1" data-bbox="446 1108 920 1140"> <tr> <td>tag(22 bit)</td> <td>sets(8 bit)</td> <td>block size(2 bit)</td> </tr> </table> now address is ABCDE8F8 its binary from is :1010 1011 1100 1101 1110 1000 1111 1000 <1010 1011 1100 1101 1110 10> <00 1111 10> <00> it is mapped to set number 62 in cache. b. 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6	tag(22 bit)	sets(8 bit)	block size(2 bit)	
tag(22 bit)	sets(8 bit)	block size(2 bit)			

Total references: 20

LRU



Page faults= 10

4

a.

 Data count register gives the number of words the DMA can transfer in a cycle.

Here it is 16 bits.. so max 2^{16} words can be transferred in one cycle..

Since memory is byte addressable.. 1 word = 1 byte
so 2^{16} bytes in 1 cycle..

Now for the given file..

$$\text{File size} = 29154 \text{ KB} = 29154 \times 2^{10} \text{ B}$$

$$1 \text{ cycle} \rightarrow \text{DMA transfers } 2^{16} \text{ B}$$

i.e

$$1 \text{ B transferred by DMA} \rightarrow \frac{1}{2^{16}} \text{ cycles.}$$

Now, for full file of size 29154 KB,

$$\text{Minimum number of cycles} = \frac{(29154 \times 2^{10} \text{ B})}{2^{16}} = 455.53$$

But number of cycles is asked so $455.53 \rightarrow 456$.

b. i) destination initiated Handshaking – When source initiates the data transfer process. It consists of signals: DATA VALID: if ON tells data on the data bus is

valid otherwise invalid. DATA ACCEPTED: if ON tells data is accepted otherwise not accepted.

ii) destination initiated strobe I/O

In many applications, valid data is present on an external device only at a certain time, so it must be read in at that time.

E.g. the ASCII-encoded keyboard. When a key is pressed, circuitry on the keyboard sends out the ASCII code for the pressed key on eight parallel data lines, and then sends out a strobe signal on another line to indicate that valid data is present on the eight data lines.

The sending device, such as a keyboard, outputs a parallel data on the data lines, and then outputs an STB signal to let you know that valid data is present. For low rates of data transfer, such as from a keyboard to a MP, a simple strobe transfer works well. However, for higher speed data transfer, this method does not work because there is no signal which tells the sending device when it is safe to send the next data byte.

5

a)

Big Endian:

Address:			0x100	0x101	0x102	0x103		
Value:			01	23	45	67		

Little Endian:

Address:			0x100	0x101	0x102	0x103		
Value:			67	45	23	01		

b)

Memory mapped IO allows peripheral devices to be accessed by mapping their control registers and data buffers to specific memory addresses. However, the address space available for memory mapped IO is limited by the size of the memory address space that the system can support.

For example, if a system has a 32-bit address bus, it can support a maximum of 2^{32} addressable locations, which correspond to 4 GB of memory address space. If a significant portion of this address space is already allocated to system memory, there may not be enough address space left to map all the peripheral devices in the system.

Additionally, some peripheral devices may require a large amount of address space for their control registers and buffers. For example, graphics cards or network controllers may have a large number of registers that need to be accessed for efficient operation. If each of these devices requires a significant portion of the available address space, it may not be possible to map all of them



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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.: 238LE0710

SLOT: C1 + TC1

Programme Name & Branch : B. Tech Computer Science and Engineering
 Course Code and Course Name : BCSE205L and Computer Architecture and Organization
 Faculty Name(s) : Common to all
 Class Number(s) : VL2024250501363, VL2024250501365, VL2024250501379, VL2024250501382, VL2024250501386, VL2024250501389, VL2024250501393, VL2024250501395, VL2024250501403, VL2024250501417, VL2024250501419, VL2024250501421, VL2024250501423, VL2024250501425, VL2024250501427, VL2024250501431, VL2024250501435, VL2024250501441, VL2024250501445, VL2024250501449, VL2024250501454, VL2024250501476

Date of Examination : 18-03-2025

Exam Duration : 90 minutes

Maximum Marks: 50

General instruction(s):

- Answer All Questions
- CO1-Differentiate Von Neumann, Harvard, and CISC and RISC architectures. Analyze the performance of machine with different capabilities. Recognize different instruction formats and addressing modes. Validate efficient algorithm for fixed point and floating-point arithmetic operations.
- CO2- Explain the importance of hierarchical memory organization. Able to construct larger memories. Analyse and suggest efficient cache mapping technique and replacement algorithms for given design requirements. Demonstrate hamming code for error detection and correction.
- CO3-Understand the need for an interface. Compare and contrast memory mapping and IO mapping techniques. Describe and differentiate different modes of data transfer. Appraise the synchronous and asynchronous bus for performance and arbitration.

Q. No	Question	M	CO	BL												
1.	a) Computer A runs a program in 40 seconds with a 2 GHz clock. Computer B is designed to run the same program in 20 seconds but requires twice the clock cycles of Computer A. Analyse how the relationship between execution time, clock cycles, and clock rate influences the performance of Computer B compared to Computer A. Discuss the trade-offs involved in increasing clock rates versus the number of clock cycles required.	5	CO1	BL4												
	b) As part of enhancing the functionality of an architecture, it is necessary to incorporate a complex instruction into its instruction set. Considering the trade-offs involved, analyse whether a hardwired or microprogrammed control approach is more suitable for implementing this update. Provide a detailed justification for your recommendation, supported by an explanatory diagram.	5														
2.	Evaluate the given expression $A = (B + C) * (D - E)$ and compute the memory to encode, memory to store, memory traffic for 3, 2, 1, and 0 address machines. Assume the following: Addresses are 24 bits, Data values are 24 bits, Opcodes are 8 bits, and Word length is 8 bits. Note: Specify the answer in this format.	10	CO1	BL3												
	<table><tr><th>1</th><th>Memory to store</th><th>Memory to Encode</th><th>Memory to Fetch</th><th>Memory to Execute</th><th>Memory Traffic</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	1	Memory to store	Memory to Encode	Memory to Fetch	Memory to Execute	Memory Traffic									
1	Memory to store	Memory to Encode	Memory to Fetch	Memory to Execute	Memory Traffic											
3.	a) Consider a machine with a byte addressable main memory of 2^{20} bytes and block size of 8 bytes. Assume that a direct mapped cache	4	CO2	BL4												

**VIT**Vellore Institute of Technology
(Chartered by the University Grants Commission, India)**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025**SLOT: C1 + TC1**

WINTER SEMESTER

	consisting of 32 lines is used with this machine. i) How is a 20-bit memory address divided into tag, line number, and byte number? ii) Into what line would bytes with each of the following addresses be stored? 1) 1101 0000 0001 1101 0001 2) 1010 1010 1010 1010 0010																					
	b) Draw the circuit for ROM memory which can hold 2×2^{10} words with word size 64 using an integrated chip of 2^{10} words with word size of 32. Give the range of addresses for each of the ICs in the circuit.	6																				
4.	a) Using a diagram, demonstrate how direct mapping organizes and stores the given sequence of memory blocks in a cache. Assume that 4-line cache memory is used. (Use the same sequence in part b) b) Apply the block replacement algorithm (FIFO and Optimal) on the sequence of block references provided, considering a fully associative cache of size 128 B with block size 32 bytes. The sequence is as follows: 5, 7, 7, 4, 1, 5, 0, 7, 7, 0, 13, 6, 5, 1, 12, 5 Summarize the detail <table border="1" style="width: 100%;"><thead><tr><th>Algorithm</th><th>FIFO</th><th>Optimal</th></tr></thead><tbody><tr><td>Number of hits</td><td></td><td></td></tr><tr><td>Number of misses</td><td></td><td></td></tr><tr><td>Total number of references</td><td></td><td></td></tr><tr><td>Hit Ratio</td><td></td><td></td></tr><tr><td>Miss Ratio</td><td></td><td></td></tr></tbody></table>	Algorithm	FIFO	Optimal	Number of hits			Number of misses			Total number of references			Hit Ratio			Miss Ratio			3	CO2	BL3
Algorithm	FIFO	Optimal																				
Number of hits																						
Number of misses																						
Total number of references																						
Hit Ratio																						
Miss Ratio																						
		7																				
5.	a) Analyse the communication challenges that might arise when a CPU interacts with an I/O device like a printer. Propose a solution involving an intermediary system to address these challenges and create a diagram to illustrate your proposed solution. b) Explain how organizing memory into four modules can improve the performance of a memory system with a 16-bit address space. Use sketches to demonstrate the concepts of lower-order and higher-order interleaving. Determine which memory module would store the word at address 0xA2C8 for both schemes and justify your reasoning.	5	CO3	BL2																		
		5																				

School of Computer Science and Engineering

Winter Semester 2023-24

Continuous Assessment Test – II

SLOT: D2-TD2

Programme Name & Branch: BCB, BCE, BCI, BCT, BDS, BKT

Course Name & Code: Computer Architecture and Organization- BCSE205L

Class Number (s): Common to all

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer all five questions

Q. No.	Question	Max Marks
1.	A program runs in 10 seconds on computer X with 2 GHz clock. Assume that the Computer Y runs the same program in 6 seconds, then compute the following: i) What is the number of CPU cycles on computer X? ii) If computer Y requires 10% more cycles to execute program when compared to computer X, then what is the clock rate for computer Y? iii) Assume the computer Z is running with clock rate of Y and the number of clock cycles as X, then find the CPU time of Computer Z. iv) Identify the fastest and slowest machine among X, Y and Z. v) Assume that the computer A executes 3 million instructions in the CPU time of the fastest Computer among X, Y and Z, determine MIPS. Solution: i) CPU cycles on computer X = $10 \text{ sec} \times 2 \times 10^9 \text{ cycles/s} = 20 \times 10^9 \text{ cycles}$ ii) CPU cycles on computer Y = $1.1 \times 20 \times 10^9 = 22 \times 10^9 \text{ cycles}$ Clock rate for computer Y = $22 \times 10^9 \text{ cycles} / 6 \text{ sec} = 3.67 \text{ GHz}$ iii) Clock rate of computer Z = $3.67 \text{ GHz} = 3.67 \times 10^9$ No. of clock cycles of Z = $20 \times 10^9 \text{ cycles}$ CPU time of computer Z = $20 \times 10^9 \text{ cycles} / 3.67 \times 10^9 = 5.44 \text{ s}$ v) CPU time of X = 10 s CPU time of Y = 6 s CPU time of Z = 5.44 s Therefore, faster machine is, Computer Z slower machine is, Computer X vi) Instruction count of computer Z = 3 million MIPS rating = $3 / 5.44 = 0.551470 \text{ s}$	10

2. A computer employs RAM chips of 1024×8 and ROM chips of 1024×8 . The computer system needs $2k \times 8$ bytes of RAM, $2k \times 8$ bytes of ROM, and four interface units with four registers in each. A memory-mapped I/O configuration is used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

- How many RAM and ROM chips are needed?
- Draw a memory address map for the system.
- Give the address range in hexadecimal for RAM, ROM, and interface.
- Develop a chip layout for the above said specifications.

Solution:

No. of Chips and Address Calculation

S/N	Memory	N x W	N' x W'	P	Q	P'Q'	x	y	z	Total
1	RAM	1024×8	2048×8	2	1	2	10	1	2	13
2	ROM	1024×8	2048×8	2	1	2	10	1	2	13
3	Interface	4		4	1	4	2	2	2	6

a) No. of RAM chips are needed $(p \times q) = 2$
and No. of ROM chips are needed : 2

b) Memory Address Map:

Component	Hexadecimal Address	Address Bus
		From To
		13 12 11 10 9 8 7 6 5 4 3 2 1 0
RAM 1	0000 0FFF	0 0 0 0 x x x x x x x x x x

Interface 2	1200	1203	1	0	0	1					x	x
Interface 3	1400	1403	1	0	1	0					x	x
Interface 4	1600	1603	1	0	1	1					x	x

Design Diagram (2m)

Solution:

1

Therefore, memory references will be present in the cache are:

72, 12, 1, 9, 26, 14, 7, 19

Hit Ratio = 10 / 30 = 0.33 %

Hit Ratio = 10 / 30 = 0.33 %

Miss Ratio = 0.67 %

a) A hard disk with using DMA. The se	
--	--

Solution:
 DMA transfer rate = $16 \text{ MBps} = 16 \times 10^6 = 10^7$ bytes per second
 Processor frequency = $f = 600 \text{ MHz} = 600 \times 10^6 = 6 \times 10^8 \text{ Hz}$
 No. of cycles used by CPU = $300 + 900 = 1200$
 Block size = $70 \text{ KCB} = 70 \times 10^3 \times 8 = 560 \text{ bytes}$.

Solution:

Processor frequency = $f = 600 \text{ MHz} = 600 \times 10^6 = 6 \times 10^8 \text{ Hz}$

Block size = 20KB = $20 \times 10^3 = 2 \times 10^4$ bytes

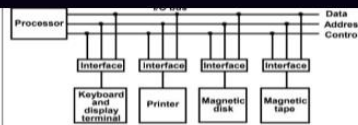
$$\text{Time consumed by CPU} = \frac{\text{No. of cycles used by}}{1000 \times 10^6} \times 10^6 \text{ s}$$
$$\text{Time consumed by DMA} = \text{Block size} / \text{DMA transfer rate}$$
$$= 2 \times 10^{-7} / 10 = 2 \times 10^{-8} \text{ sec}$$

b) Discuss the functionality of interfaces in I/O subsystem with neat diagram.

Key points:

- Each peripheral has an interface module to get connected with the computer system.
- Each Interface decodes the address and control received from the I/O bus, interprets them for peripherals and provides signals for the peripheral controller.
- It is also synchronizes the data flow and supervises the transfer between peripheral and processor.
- Each peripheral has its own controller.

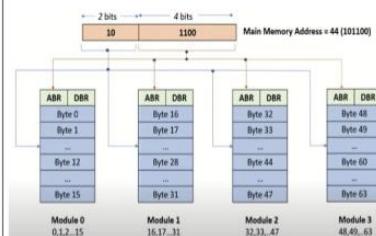
Diagram



Note: Expected to explain

5. a) Consider a byte addressable memory of 64 bytes with 4 way high-order interleaved. Determine the following:
- If the CPU generates 101100 main memory address, then identify the bits required to select the module and the bits required to select the specific location in the selected module. Also mention the selected module and specific location in the selected module for the above mentioned address.
 - Sketch the structure of the above interleaved main memory.
- Solution:**
- Given, 4 way interleaving = 4 modules = 2^2
 Modules are labeled as m0, m1, m2, and m3
- Bits required to select the module = 2
- For High order interleaving, high order 2 bits are required to identify each module.
 Given main memory address = 101100
 Therefore, Module bits to be used is = 10 (i.e. m2)
 Bits required to select the specific location in the selected module = 1100 (LSB bits)
 Selected module is m2 (module 2)
 Specific location in the selected module : Byte 44 (11th location in module 2)

ii) Structure



b) Explain different phases of instruction cycle with neat sketch. Describe the various CPU registers involved for the completion of each instruction execution.

Different phases are:

- Fetch stage
- Decode stage
- Execute stage

Diagram

Instruction Cycle State Diagram



Note: Expected to explain each phase

(Op)

Reg. No.:

Name :

**VIT**

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – June 2023

Programme	: B.Tech (ECM)	Semester	: Fall Inter 2022-23
Course	: Computer Architecture and Organization	Code	: BCSE205L
Faculty	: Dr. Sindhuja M	Slot	: A2+TA2
	Dr. Sunil Kumar Pradhan	Class Nbr.	: CH2022232500336
	Dr. Balakrishnan R		CH2022232500337
	Mr. Ajeyprasaath Kb		CH2022232500338
			CH2022232500340
Time	: 90 Minutes	Max. Marks	: 50

Answer ALL the questionsQ.No. Sub.
Sec.

Questions

Marks

Consider a 2-level memory hierarchy with separate instruction and data caches in level 1, and main memory in level 2. Figure 1 for the reference.

10

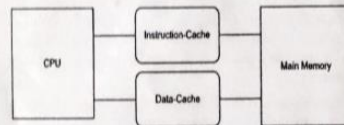


Figure 1

1.

The following parameters are given:

- The clock cycle time is 2ns
- The miss penalty is 12 clock cycles (for both read and write)
- 2% of instructions are not found in instruction cache
- 5% of data references are not found in data cache
- 25% of the total memory accesses are for data
- Cache access time (including hit detection) is 1 clock cycle

Find average access time.

2.

How many 256 x 4 RAM chips are needed to construct a memory chip of 2048x8?
Show the corresponding interconnection diagram.

10

- Consider a direct mapped cache of size 4 KB with block size 1024 bytes. The size of main memory is 1MB. Find-
3.
 1. Number of Bits in Physical Address [2 marks]
 2. Number of Bits in Block Offset [2 marks]
 3. Number of bits in tag [3 marks]
 4. Tag directory size [3 marks]
- 10
- (a) On a non-pipelined sequential processor, the following program segment, that is part of the Interrupt Service Routine (ISR), is given to transfer 500 bytes from an I/O device to memory.
- Initialize the memory address register*
Initialize word count register to 500
Loop: Load a byte from device
Store in memory at address given by address register
Increment the memory address register
Decrement the word count register
If count != 0 goto Loop
4. Assume that each statement in the program is equivalent to a machine instruction that takes 1 clock cycle to execute if it is a non LOAD/STORE instruction. The LOAD/STORE instructions take 2 clock cycles to execute. The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes 2 clock cycles to transfer one byte of data from the device to the memory.
- 10
- What is the approximate speedup when the DMA controller-based design is used in place of the interrupt driven approach? [8 marks]
- (b) Suppose that a disk is rotating at a speed of 10,000 rpm, and there are 120 Kbytes of data recorded in every track. Once the disk head reaches the desired track, what is the sustained data transfer rate in Mbytes/sec? [2 marks]
- The size of the word count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 Kbytes from disk to main memory. The memory is byte addressable. What is the minimum number of times the DMA controller needs to get control of the system bus from the processor to transfer the file from disk to main memory? [5 marks]
- (a)
5. Assume 1 kb = 1024 bytes.
- 10
- (b) A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much % will the CPU be slowed down because of the DMA transfer? [5 marks]



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT:C2+TC2

Programme Name and Branch : B.Tech CSE

Course Code and Course Name : Computer Architecture and Organization- BCSE205L

Q.No	Question	M	CO	BL
1.	Write the assembly code for the given arithmetic statement $W=(X+Y)/Z$ using 0,1,2 and 3 address instructions. Compute memory traffic, memory required to store and encode instructions for 0,1,2 and 3 address machines. Assuming the following fields: addresses - 32 bits, data values -32 bits, opcode - 8 bits and word length - 4 bytes.	10	1	3

	3-address	Memory Access Fetch	Memory Access Execute	Memory Traffic	Memory To Store	Memory To Encode
3-Address	ADD W, X, Y DIV W, W, Z	4 4	3 3	7 7	4*2=8 4*2=8	1*(3*2)=3 1*(3*2)=3
Total				14	16 Bytes	14 Bytes
2-Address	MOV W, X ADD W, Y DIV W, Z	3 3 3	2 3 3	5 5 6	3*2=6 3*2=6 3*2=6	1*(3*2)=3 1*(3*2)=3 1*(3*2)=3
Total				17	18 Bytes	15 Bytes
1-Address	LOAD X ADD Y DIV Z STORE W	2 2 2 2	1 1 1 1	3 3 3 3	2*2=4 2*2=4 2*2=4 2*2=4	1*(3*2)=3 1*(3*2)=3 1*(3*2)=3 1*(3*2)=3
Total				12	16 Bytes	12 Bytes
0-Address	PUSH X PUSH Y ADD PUSH Z DIV POP W	2 2 1 2 1 2	1 1 0 1 0 1	3 3 1 3 1 3	2*2=4 2*2=4 2*1=2 2*2=4 2*1=2 2*2=4	1*(3*2)=3 1*(3*2)=3 1 1*(3*2)=3 1 1*(3*2)=3
Total				14	20 Bytes	14 Bytes

2. Consider a processor with three instruction classes X, Y and Z, with the corresponding CPI values being 3, 2 and 1 respectively. The processor runs at a clock rate of 2 GHz. For a given program, two compilers produce the following executed instruction counts. Compare the performance of both compilers by computing MIPS and execution time.

Instruction Count (in millions)		
	X	Y
Compiler 1	4	6
Compiler 2	3	11

Answer

For compiler 1:

$$CPI_1 = \frac{3 \times 4 + 2 \times 6 + 1 \times 11}{4+6+11} = 28 / 15 = 1.87$$

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$$MIPS_1 = \frac{2000 \text{ MHz}}{1.87} = 1069.52 \text{ MIPS}$$

$$CPU \text{ Execution Time}_1 = \frac{(4+6+11) \times 10^6 \times 1.87}{10^9} = 0.028 \text{ sec}$$

For compiler 2:

$$CPI_2 = \frac{3 \times 3 + 2 \times 11 + 1 \times 11}{3+11+11} = 34 / 21 = 1.62$$

$$MIPS_2 = \frac{2000 \text{ MHz}}{1.62} = 1234.57 \text{ MIPS}$$

$$CPU \text{ Execution Time}_2 = \frac{(3+11+11) \times 10^6 \times 1.62}{10^9} = 0.034 \text{ sec}$$

An instruction is stored at location 1000 with its address field at location 1001: the address field has a value of 4000. Address Location 4000 contains 1500. A processor register R1 contains the number 1000 and Index register XR contains the number 200. Evaluate the effective address of the following addressing modes: Direct, Indirect, Relative, Indexed, Register Indirect Addressing Mode.

Answer

- Direct addressing: EA: Address Field = 4000
- Indirect addressing: EA: (Address Field) = (4000) = 1500
- Relative addressing: EA: PC + Address Field = 1002+4000 = 5002
- Indexed addressing: EA: Address Field + Index Register (XR) = 4200
- Register addressing: EA: R1 = 1000

3. A computer employs RAM chips of 1KB and ROM chips of 512 x 8. The computer system needs 1GB of RAM, 4M x 16 of ROM, and two interface

3. A computer employs RAM chips of 1KB and ROM chips of 512 x 8. The computer system needs 1GB of RAM, 4M x 16 of ROM, and two interface units with 256 registers each. A memory mapped I/O configuration is used. The two higher-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

a. Compute the total number of decoders needed for the above system. (4 Marks)

b. Design a memory-address map table for the above system. (4 Marks)

c. Show the chip layout for the above design. (2 Marks)

Answer

S.NO	Memory	N x W	N' x W'	P	q	P * q	x	y	z	Total
1	RAM	1KB x 8	1GB x 8	2 ¹⁰	1	2 ¹⁰	10	20	2	32
2	ROM	512 x 8	4MB x 16	2 ¹³	2	2 ¹⁴	9	14	2	25
3	Interface	256	512	2	1	2	8	1	2	11

Hex Address		11-30	29-16	16-10	9	8	T-6
From	To						
RAM1	00000000	000000FF	00	00.00	00.00	x	x x.x
RAM2	00000200	000007FF	00	00.00	00.01	x	x x.x

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<table><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>RAM2</td><td>00000000</td><td>000000FF</td><td>00</td><td>11.11</td><td>11.11</td><td>x</td><td>x</td><td>x</td><td>x.x</td></tr><tr><td>ROM1.1</td><td>00000000</td><td>000000FF</td><td>01</td><td>00.00</td><td></td><td>x</td><td>x</td><td>x</td><td>x.x</td></tr><tr><td>ROM1.2</td><td>00000000</td><td>000000FF</td><td>01</td><td>00.01</td><td></td><td>x</td><td>x</td><td>x</td><td>x.x</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>Interface1</td><td>10000000</td><td>100000FF</td><td>10</td><td>0</td><td></td><td></td><td></td><td></td><td>x.x</td></tr><tr><td>Interface2</td><td>50000000</td><td>500000FF</td><td>10</td><td>1</td><td></td><td></td><td></td><td></td><td>x.x</td></tr></table>			...	...	...	...	...	...	...	...	...	...	RAM2	00000000	000000FF	00	11.11	11.11	x	x	x	x.x	ROM1.1	00000000	000000FF	01	00.00		x	x	x	x.x	ROM1.2	00000000	000000FF	01	00.01		x	x	x	x.x	...	...	...	...	...	...	...	...	...	...	Interface1	10000000	100000FF	10	0					x.x	Interface2	50000000	500000FF	10	1					x.x		
...	...	...	...	...	...	...	...	...	...																																																																	
RAM2	00000000	000000FF	00	11.11	11.11	x	x	x	x.x																																																																	
ROM1.1	00000000	000000FF	01	00.00		x	x	x	x.x																																																																	
ROM1.2	00000000	000000FF	01	00.01		x	x	x	x.x																																																																	
...	...	...	...	...	...	...	...	...	...																																																																	
Interface1	10000000	100000FF	10	0					x.x																																																																	
Interface2	50000000	500000FF	10	1					x.x																																																																	
4.	Consider a computer with the following characteristics: total of 1Mbyte of main memory; word size of 1 byte; block size of 16 bytes; and cache size of 64 Kbytes. a. For the main memory addresses of F0010, 01234, and CABBE, give the corresponding tag, cache line address, and word offsets for a direct-mapped cache. (3 Marks) b. Give any two main memory addresses with different tags that map to the same cache slot for a direct-mapped cache. (2 Marks) c. For the main memory addresses of F0010 and CABBE, give the corresponding tag and offset values for a fully associative cache. (2 Marks) d. For the main memory addresses of F0010 and CABBE, give the corresponding tag, cache set, and offset values for a two-way and four-way set-associative cache. (3 Marks) Answer a. Direct Mapping memory addresses: Tag (4 bits) Index(12 bits) Offset(4 bits) F0010 = 1111 0000 0000 0001 0000 • Offset (4 bits) = 0000 = 0 (Hex) • Index (12 bits) = 0000 0000 0001 = 1 (Hex) • Tag (4 bits) = 1111 = F (Hex) 01234 = 0000 0001 0010 0011 0100 • Offset (4 bits) = 0100 • Index (12 bits) = 0010 0011 0100 • Tag (4 bits) = 0000 CABBE = 1100 1010 1011 1011 1110 • Offset (4 bits) = 1110 • Index (12 bits) = 1010 1011 1011			2	3																																																																					
				10																																																																						

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• Tag (4 bits) = 1100		
b.	01234 = 0000 0001 0010 0011 0100 Index = 0x234 Tag = 0 81234 = 1000 0001 0010 0011 0100 Index = 0x234 Tag = 8	
c. Fully Associative Cache	F0010 Tag = 1111 0000 0000 0001 Offset = 0000 (0) CABBE Tag = 1100 1010 1011 1011 Offset = 1110 (E)	
d. Set-associative	2-way addressing: Tag (5 bits) Index(11 bits) Offset(4 bits) 4-way addressing: Tag (6 bits) Index(10 bits) Offset(4 bits) F0010: 1111 0000 0000 0001 0000 2-Way: Tag = 11110 Set Index = 0000 0000 000 Offset = 0000 4-Way: Tag = 111100 Set Index = 0000 0000 00 Offset = 0000 CABBE: 1100 1010 1011 1011 1110 2-Way: Tag = 11001 Set Index = 0101 0111 011 Offset = 1110 4-Way: Tag = 110010 Set Index = 1010 1011 10 Offset = 1110	
5.	Explain how I/O Controller manages communication between the CPU and peripheral devices in detail with neat diagrams. Answer An I/O Controller is a specialized hardware component that acts as a bridge between the CPU, memory, and peripheral devices (such as printers, keyboards,	3 2 4

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hard drives, and network interfaces). It manages communication, controls data transfers, and offloads processing from the CPU.		
The I/O Controller performs several key functions:		
- Interface Management: Connects peripheral devices to the CPU. - Data Transfer Control: Manages the movement of data between memory and I/O devices. - Synchronization: Ensures that slow peripheral devices do not affect CPU performance. - Interrupt Handling: Sends interrupts to the CPU when a device is ready. - Error Detection & Correction: Ensures reliable data transmission.		
The I/O Controller manages communication between the CPU and peripheral devices in a step-by-step manner:		
Step 1: CPU Sends a Command		
The CPU sends a request (Read/Write) to the I/O Controller via the System Bus.		
The command includes:		
Device Address (to specify which device to communicate with).		
Operation Type (Read or Write).		
Memory Address (where to store or retrieve data).		



Continuous Assessment Test (CAT) – II October 2025

Programme	: B. Tech (CSE and all Specializations)	Semester	: FALL 25-26
Course Code & Course Title	: BCSE205L-Computer Architecture and Organization	Slot	: D1 +TD1
Faculty	: Dr. MANJULA V Dr. KAJA MOHIDEEN A Dr. ASWIGA Dr. REVATHI A R Dr. PADMA J	Class Number	: CH2025260100953 CH2025260100954 CH2025260100955 CH2025260100956 CH2025260100957
Duration	: 90 Minutes	Max. Mark	: 50 Marks

General Instructions

- Write only your registration number on the question paper in the box provided, and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only a non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level												
1	a	Consider three different processors, P1, P2, and P3, that execute the same instruction set. <table><tr><th>Processor</th><th>Clock rate</th><th>CPI</th></tr><tr><td>P1</td><td>3 GHz</td><td>1.5</td></tr><tr><td>P2</td><td>2.5 GHz</td><td>1.0</td></tr><tr><td>P3</td><td>4 GHz</td><td>2.2</td></tr></table> i. Which processor has the highest performance? (4 Marks) ii. If each processor executes a program in 10 seconds, find the number of cycles and the number of instructions. (3 Marks) iii. When you try to reduce the execution time by 30% but this leads to an increase of 20% in the CPI. Determine the clock rate required to achieve this time reduction. (3 Marks)	Processor	Clock rate	CPI	P1	3 GHz	1.5	P2	2.5 GHz	1.0	P3	4 GHz	2.2	10	2	K3
Processor	Clock rate	CPI															
P1	3 GHz	1.5															
P2	2.5 GHz	1.0															
P3	4 GHz	2.2															
1	b	i. Compare the Hardwired and Microprogrammed control units in terms of their suitability for implementing 30 additional instructions in a CPU. Which control path approach is easier to add new instructions to? Justify your answer. (2 Marks) ii. Discuss the key differences between the two approaches. (3 Marks)	05	2	K3												
2		A fully associative cache has 4 blocks (block size = 1 word). The cache is initially empty. The memory reference string (decimal addresses) is 10, 20, 30, 40, <u>10</u>, 20, 10, 50, 20, 30, 10, 20, 30, 40, 10 i. Simulate the cache contents for FIFO replacement policy. (2 Marks) ii. Simulate the cache contents for LRU replacement policy. (2 Marks) iii. Compute the hit ratio and miss ratio for both FIFO and LRU. (2 Marks)	10	2	K3												

		iv. Compute compulsory miss and conflict miss for both FIFO and LRU. (2 Marks)			
		v. Which policy gives a better hit ratio for this reference string? Explain why the results differ. (2 Marks)			
3		A microcomputer system requires a total of $512K \times 32$ byte-addressable RAM. RAM chips available are $64K \times 8$ each. Design an interleaved memory system based on the following requirements: i. The total number of RAM chips required; the number of RAM banks required to support memory interleaving. Also, calculate the number of RAM chips per bank. (3 Marks) ii. How many address lines are needed to select the bank, and decoder logic with appropriate illustrations (3 Marks) iii. If the system is expanded to $1M \times 32$ RAM using the same chips, how will the number of banks and chips per bank change? (4 Marks)	10	2	K3
4	a	A critical goal in modern embedded systems design is maximizing CPU throughput and minimizing power consumption during high-volume data transfers. Analyse the architectural role of a Direct Memory Access Controller within a Microcontroller Unit by addressing the following i. Describe the three key registers the CPU must program into the DMA Controller to initiate a transfer between a peripheral and RAM. (5 Marks) ii. Recommend which mode of transfer is preferred for time-critical, high-bandwidth communication. Justify your choice. (5 Marks)	10	2	
	b	Analyze the two fundamental I/O communication models used below for interfacing a microcontroller with a sensor and a display. The sensor and display are mapped to the microcontroller's memory address space for access. The sensor and output device are accessed via dedicated ports from the microcontroller's memory space i. Identify the communication mechanism for each method described above. (2 Marks) ii. Write instructions for the following operations within each model and recommend the best approach for modern embedded systems with a technical justification. • Reading data from the sensor. • Writing data to the display. (3 Marks)	05	2	K3
*****All the best *****					